ON THE NONEXISTENCE OF A CYCLIC $(8835, 4417, 2208)$-DIFFERENCE SET

Abstract. We prove nonexistence of a cyclic difference set with parameters $(8835, 4417, 2208)$. The argument uses the multiplier 47, forced residue profiles modulo 3, 5, and 15, and a final orbit-count contradiction on the block of elements congruent to 5 or 10 modulo 15. This removes one of the named small open cyclic Hadamard cases recorded by Baumert and Gordon.

1. Introduction

Baumert and Gordon list the parameter set $(8835, 4417, 2208)$ among the open cyclic Hadamard cases up to 10000 [1], and Gordon's La Jolla status slides continue to display 8835 among the small unresolved cyclic Hadamard parameters [2]. We show that the case is impossible.

Theorem 1. *There is no cyclic* $(8835, 4417, 2208)$*-difference set.*

Proof. Assume for contradiction that $D \subseteq G = C_{8835}$ is a $(8835, 4417, 2208)$-difference set. Set

$$n = k - \lambda = 4417 - 2208 = 2209 = 47^2.$$

By the standard prime-power multiplier theorem, 47 is a multiplier. Since

$$\gcd(47 - 1, 8835) = \gcd(46, 8835) = 1,$$

we may translate D and assume

$$47D = D.$$

Because $47 \equiv 2 \pmod{m}$ for $m \in \{3, 5, 15\}$, the residue counts of D modulo each of these moduli are constant on the multiplication-by-2 orbits.

Step 1: the profile modulo 3. Let N_i be the number of elements of D congruent to i modulo 3. Since multiplication by 2 swaps the nonzero classes of $\mathbb{Z}/3\mathbb{Z}$, we have $N_1 = N_2$. For a nontrivial character of $\mathbb{Z}/3\mathbb{Z}$ with value ω on 1, the lifted character sum is

$$N_0 + N_1\omega + N_2\omega^2 = N_0 - N_1.$$

Every nonprincipal character value has modulus 47, so $|N_0 - N_1| = 47$. Together with $N_0 + 2N_1 = 4417$, this forces

$$(N_0, N_1, N_2) = (1441, 1488, 1488).$$

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Step 2: the profile modulo 5. Let M_j be the number of elements of D congruent to j modulo 5. Multiplication by 2 acts transitively on the four nonzero residue classes, so

$$M_1 = M_2 = M_3 = M_4.$$

For a nontrivial order-5 character with primitive value ζ_5, the lifted character sum is

$$M_0 + M_1(\zeta_5 + \zeta_5^2 + \zeta_5^3 + \zeta_5^4) = M_0 - M_1.$$

Again the modulus is 47, so $|M_0 - M_1| = 47$. Since $M_0 + 4M_1 = 4417$, we obtain

$$(M_0, M_1, M_2, M_3, M_4) = (921, 874, 874, 874, 874).$$

Step 3: the profile modulo 15. Multiplication by 2 on $\mathbb{Z}/15\mathbb{Z}$ has orbits

$$\{0\}, \quad \{5, 10\}, \quad \{3, 6, 9, 12\}, \quad \{1, 2, 4, 8\}, \quad \{7, 11, 13, 14\}.$$

Let the common counts on these five orbits be

$$a,\ b,\ c,\ d,\ e$$

respectively. Comparing with the profiles modulo 3 and 5 found above yields

$$a + 4c = 1441, \tag{1}$$

$$a + 2b = 921, \tag{2}$$

$$c + d + e = 874, \tag{3}$$

$$b + 2d + 2e = 1488. \tag{4}$$

Now choose a primitive character of $\mathbb{Z}/15\mathbb{Z}$ with value ζ on 1 and set

$$U = \zeta + \zeta^2 + \zeta^4 + \zeta^8.$$

Then the sum on the complementary unit orbit is $\overline{U}$, and one has

$$U + \overline{U} = 1, \qquad U = \frac{1 \pm i\sqrt{15}}{2}.$$

Therefore the lifted character sum is

$$A_{15} = a - b - c + dU + e\overline{U},$$

and $|A_{15}| = 47$. Using (1)–(4), we obtain

$$a = 1441 - 4c, \qquad b = 2c - 260, \qquad d + e = 874 - c,$$

hence

$$A_{15} = \frac{4276 - 15c}{2} + \frac{(d - e)i\sqrt{15}}{2}.$$

The equation $|A_{15}|^2 = 47^2$ becomes

$$(4276 - 15c)^2 + 15(d - e)^2 = 8836.$$

Since $|4276 - 15c| \leq 94$ and $4276 - 15c \equiv 1 \pmod{15}$, there are only finitely many candidates for the first term. A direct check shows that the unique possibility is

$$4276 - 15c = -14, \qquad d - e = \pm 24.$$

Thus

$$c = 286, \qquad b = 312, \qquad a = 297, \qquad \{d, e\} = \{306, 282\}.$$

In particular, the block

$$B = \{x \in \mathbb{Z}/8835\mathbb{Z} : x \equiv 5 \text{ or } 10 \pmod{15}\}$$

contains exactly

$$2b = 624$$

elements of D.

Step 4: the final orbit obstruction. Because $47D = D$, the set D is a union of full 47-orbits in C_{8835}. Every element of B has additive order one of

$$3,\ 57,\ 93,\ 1767,$$

so the possible 47-orbit sizes inside B are

$$\mathrm{ord}_3(47) = 2, \qquad \mathrm{ord}_{57}(47) = 18,$$
$$\mathrm{ord}_{93}(47) = 10, \qquad \mathrm{ord}_{1767}(47) = 90.$$

The numbers of such orbits available in the whole group are

$$\frac{\varphi(3)}{2} = 1, \qquad \frac{\varphi(57)}{18} = 2, \qquad \frac{\varphi(93)}{10} = 6, \qquad \frac{\varphi(1767)}{90} = 12.$$

Hence the total 624 points of $D \cap B$ would have to satisfy

$$624 = 2\alpha + 18\beta + 10\gamma + 90\delta$$

with

$$0 \le \alpha \le 1, \qquad 0 \le \beta \le 2, \qquad 0 \le \gamma \le 6, \qquad 0 \le \delta \le 12.$$

After dividing by 2,

$$312 = \alpha + 9\beta + 5\gamma + 45\delta.$$

If $\delta \le 4$, then the right-hand side is at most

$$1 + 18 + 30 + 180 = 229,$$

so $\delta \ge 5$. If $\delta = 5$, the remainder is

$$87 = \alpha + 9\beta + 5\gamma,$$

but the right-hand side is at most 49. If $\delta = 6$, the remainder is

$$42 = \alpha + 9\beta + 5\gamma,$$

which is impossible for $\alpha \in \{0, 1\}$, $0 \le \beta \le 2$, and $0 \le \gamma \le 6$. If $\delta \ge 7$, then $45\delta > 312$. This contradiction proves that no cyclic $(8835, 4417, 2208)$-difference set exists. □

Remark 2. The proof isolates a rigid mod-15 profile and then shows that the required $\{5, 10\} \pmod{15}$ block cannot be assembled from full 47-multiplier orbits. No search-heavy construction check is needed.

References

[1] Leonard D. Baumert and Daniel M. Gordon, *On the existence of cyclic difference sets with small parameters*, Fields Inst. Commun. **41** (2004), 29–40.

[2] Daniel M. Gordon, *The La Jolla Difference Set Repository*, ArasuFest talk slides, August 3, 2019.